



The new 34nm MLC flash based SSD drives from Intel get a firmware upgrade to resolve data corruption issues face by some customers

Electronics gets Bio shot!

Lawrence Livermore National Laboratory scientists are claiming that Bionanoelectronic devices can speed up electronics and even processing

the memory and bus subsystem is to a processor's performance. It also underlines the importance of avoiding a bottleneck. AMD's Athlon 7850BE was the only "black edition" chip in this group, meaning its multiplier is 'upward unlocked'.

Most processors have multipliers that can scale down, but never scale above their



AMD Athlon X2 250

maximum value. These black edition CPUs are intended for overclockers. Our Best Buy award goes to the Athlon II X2 250 from AMD, a solid performer that makes an excellent buy at a price point of Rs. 4,150.

Processors between Rs. 5,001 and Rs. 12,500 - Shooting from the hip

For many people, this wide range represents the sweet spot - this is where processors start getting interesting and the term *future proofing* starts to emerge. Entry-level quad and tri-cores make their presence felt and while some still feel anything more than two cores is overkill, not to mention underutilised, these processors are most certainly the future. In fact, both companies have announced their intentions to make six-core processors available by next year. Parallelism is the shape of things to come, graphics cards have already gone massively parallel and processors aren't lagging far behind. The CPUs

in this category are certainly a diverse bunch and a simple price to price comparison won't bring out every aspect of their abilities. Some are better suited to a particular set of tasks while others excel at others.

Faster clock speeds used to be the order of the day for gamers, but tread with caution here, because the newer crop of games are slowly becoming multi-threaded. However, that's not to say clock speed is not important. In fact, from the many games we tried and discarded we found that *S.T.A.L.K.E.R. Clear Sky*, *Crysis Warhead* and *Far Cry 2* all respond with greater fervour to clock speeds than number of cores. In general, RTS and RPG games, because of their complex AI and AI pathfinding routines, tend to offer improved performance

for each extra core added. Most racing simulations and fps games tend to favour clock speeds - although this is not a hard and fast rule and exceptions exist.

After AMD's dominance in the previous category, this category was a little more even between processors from both brands. The cheapest CPUs are Intel's Core 2 Duo E7200 and E7400. It's also interesting to find the Core 2 Duo E8400 and the quad-core duo (Intel Q8200 and Q8400) very close to each other in terms of price. Although the former is a high clocked dual core at 3 GHz, the latter offers two extra cores at lower clock speeds, i.e. 2.33 GHz and 2.66 GHz, respectively. These processors are all priced within Rs. 200 of each other. AMD squeezes in one dual core, one tri-core and two quad cores. Of special interest is the Phenom II X4

905E, the suffix E denotes an energy efficient design and this CPU's TDP (Thermal Design Power) is only 65 watts, compared to the higher clocked Phenom II 810 with a TDP of 95 watts. Note that all these AMD CPUs bear the Phenom 2 moniker, meaning they have L3 caches, whereas any CPU with the Athlon nomenclature doesn't have an L3 cache.

Doom 3 shows its age in terms of multi-core support - the highest AMD



Intel Core 2 Quad Q8400

